

Sai Sri Harsha Kumbam

Software Development Engineer

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SUMMARY

Results-driven **Software Development Engineer** with extensive experience developing **scalable software applications, backend automation frameworks, and hardware/software integration** suites for enterprise cloud and production environments. Skilled in **Python, Java, Bash, RESTful API design, and containerized architectures (Docker, NGINX)**, alongside hands-on expertise in **server/rack-level validation (L10/L11)**, stress testing, and **network provisioning infrastructure (PXE, DHCP, VMware)**. Adept across the full **Software Development Life Cycle (SDLC)** including architecture design, **code reviews**, and **CI/CD pipelines** via **Git, Jira, and Azure DevOps** collaborating cross-functionally to optimize application performance, eliminate production bottlenecks, and maximize **system reliability**.

PROFESSIONAL EXPERIENCE

Software Development Test Engineer | Jabil Inc., KY, USA

08/2026- Present

- Architect, develop, and maintain **end-to-end test software applications**, test suites, and **automation frameworks** leveraging **Python, Java, and Bash** for enterprise **server (L10)** and **rack-level (L11)** production environments.
- Build **automated diagnostic engines** and programmatic test runners integrating **IPMI** and **Redfish REST APIs** to control hardware states, parse telemetry, and automate execution flows without manual intervention.
- Develop modular test scripts and execution wrappers to automate **hardware stress testing** across **CPUs, memory, NVMe/SSDs, and PCIe devices (GPUs, NICs)**.
- Configure and automate network deployment infrastructure using **PXE boot, DHCP Relay, NGINX, and VMware virtual machines (VMs)** to streamline unattended OS provisioning and test software staging.
- Maintain the software release lifecycle and source control via **Git**, conducting structured **code reviews** and managing builds, test deployments, and automated testing pipelines for **new features, system configurations, and framework updates**.
- Collaborate with **cross-functional teams** including hardware, manufacturing, and quality engineering managing sprint deliverables, bug tracking, and automated task workflows using **Jira** and **Azure DevOps (ADO)**.
- Troubleshoot complex software/hardware integration fallout via **root cause analysis (RCA)**, resolving runtime failures, network socket exceptions, and **firmware-level API communication bugs**.

Software Test Engineer | Quanta Computer Inc. 廣達電腦, TN, USA

05/2025- 08/2026

- Developed and maintained **test automation frameworks using Python and Bash scripting**, automating operational workflows to reduce manual effort, improve test coverage, and minimize production defects in high-volume environments.
- Built, configured, and validated **server infrastructure for cloud and AI workloads**, ensuring scalability, reliability, and alignment with client requirements.
- Implemented **configuration management and deployment automation using Ansible**, enabling consistent, scalable updates across multiple racks and reducing deployment time.
- Designed and enhanced a **real-time monitoring and data processing platform**, integrating hardware test stations, PXE systems, and databases to enable end-to-end visibility and operational insights.
- Migrated applications to a **containerized architecture using Docker and NGINX**, implementing load balancing to improve system scalability, availability, and fault tolerance.
- Developed full-stack solutions using **React, Node.js/Express, REST APIs, MySQL, and JSON-based configuration frameworks**, enabling dynamic workflows, real-time monitoring, and data visualization for hardware test operations.
- Implemented **logging, monitoring, and error-handling frameworks**, and performed **root cause analysis (RCA)** to resolve hardware and software integration issues, reducing downtime and improving system reliability.
- Automated **hardware diagnostics and log collection (BMC/BIOS, firmware-level debugging)**, accelerating issue resolution and improving cross-team efficiency.
- Analyzed large-scale production data using **SQL**, improving test capacity, yield, and overall manufacturing performance, utilized **Git for version control and CI/CD integration**, ensuring collaboration, traceability, and efficient code management.

- Developed and executed **comprehensive test plans and automated test cases** for functional, regression, performance, and integration testing across cloud-based and enterprise applications.
- Automated **UI and API tests** using Selenium, JUnit/TestNG, and Cucumber, significantly reducing manual effort and improving overall test coverage.
- Designed and implemented **automated testing frameworks** using Java, Python, and Groovy, enabling continuous quality assurance within CI/CD pipelines leveraging Jenkins and GitHub Actions.
- Performed extensive **API and microservices testing** using REST interfaces, Postman, and containerized services on Docker and Kubernetes, validating system integration and reliability.
- Collaborated with cross-functional teams including development, product management, and DevOps to troubleshoot defects, perform root cause analysis, and streamline the software delivery process.
- Executed **performance and load testing** using JMeter and LoadRunner, identifying bottlenecks and improving stability for cloud-native applications on AWS and Azure.
- Ensured **end-to-end quality assurance** in Agile/Scrum environments, contributing to scalable and maintainable software platforms across automotive and healthcare domains.

Programmer Analyst | Cognizant, Hyderabad, India**08/2021 – 01/2023**

- Developed **end-to-end automation for regression testing** using Selenium, TestNG, and Maven, reducing manual effort and accelerating web application deployments.
- Designed and executed **API and microservices test scenarios** in Postman, validating RESTful services and ensuring robust system integration during Agile sprints.
- Authored and maintained **test cases in TestLink**, improving coverage for fintech applications and supporting end-to-end validation of critical payment workflows.
- Managed the **defect lifecycle in Jira**, collaborating closely with development and QA teams to resolve issues efficiently and maintain software quality.
- Built a **data-driven testing framework using Apache POI and TestNG**, automating complex scenarios by dynamically reading/writing Excel test data.
- Partnered with **cross-functional teams** including developers, product managers, and DevOps engineers to troubleshoot defects, perform root cause analysis, and streamline testing processes.
- Contributed to **continuous integration efforts** by integrating automated tests into CI/CD pipelines using Jenkins and Git, ensuring faster feedback and higher release confidence.
- Supported **knowledge sharing and process improvements**, mentoring junior testers and documenting reusable test strategies for team efficiency.
- Ensured **compliance and quality standards** were maintained for enterprise and fintech applications, aligning with both internal QA processes and client expectations.

Software Testing Intern | Cognizant, Hyderabad, India**03/2021 – 08/2021**

- Developed and maintained **Selenium automation scripts** for multiple projects, reducing manual testing effort and improving test efficiency.
- Collaborated with development teams to **identify and resolve defects early**, preventing post-launch issues.
- Generated **reports and dashboards using Excel and Jira**, ensuring timely updates for stakeholders and improving project tracking.
- Assisted in **functional, regression, and integration testing**, gaining experience across multiple application layers.
- Supported **test case design and execution**, contributing to improved test coverage and overall software quality.

SKILLS

- **Programming & Scripting:** Python, Bash, Java, C/C++ (Foundational), Groovy, SQL
- **Automation & Frameworks:** PyTest, TestNG, Selenium WebDriver, Ansible, Maven, Apache POI, Cucumber (BDD), Cypress, Data-Driven & Hybrid Frameworks, Page Object Model (POM)
- **Hardware Validation & Stress Diagnostics:** IPMI, Redfish APIs, FIO, mprime, Linpack, ptugen, memtester, L10/L11 Server & Rack Testing, PCIe/NVMe Validation, Firmware/BMC Debugging
- **Systems, Virtualization & Networking:** Linux (Ubuntu, CentOS), VMware (ESXi/Workstation), PXE Boot Environments, DHCP / DHCP Relay, NGINX, BGP, ECMP, L2/L3 Switching & Routing
- **Full-Stack & Data:** Node.js, Express, React, MySQL, RESTful APIs, gRPC Microservices, JSON-driven Configuration, Real-Time Dashboards & Data Visualization
- **CI/CD, Cloud & DevOps:** Docker, Jenkins, GitHub Actions, Azure DevOps (ADO), Git/GitHub/GitLab, AWS, Azure, Server Provisioning Automation
- **Test Management & Observability:** Jira, TestLink, TestRail, Zephyr, Postman, REST Assured, JMeter, LoadRunner, Root Cause Analysis (RCA)
- **Methodologies & Practices:** SDLC, STLC, Agile/Scrum, Code Reviews, Sprint Planning, Cross-Functional Collaboration

CERTIFICATIONS

- Oracle Certified Associate, Java SE 8 Programmer
- **Cognizant Academy** - Test Case Creation – Simulation, Effective Test Case Creation, Test Data Preparation
- Certified Jr. Test Analyst/Quality Engineer – Selenium
- Test Analyst/Quality Engineer – JavaScript

EDUCATION

Masters in Information Systems	01/2023 – 12/2024
University of Maryland Baltimore County (UMBC), Baltimore, United States GPA: 3.70/4.00	
Bachelor of Technology in Electrical and Electronics Engineering	08/2017 – 06/2021
Geethanjali College of Engineering and Technology, Hyderabad, India GPA: 3.67/4.00	